ZYZYX AUTHORITY SYSTEM™

Human-Bound Artificial Intelligence (HBAI)

Defining Authority, Control, and Responsibility in Automated Systems

Authority must be present — actively — at the moment of execution. This is not an optimisation. It is the minimum condition for legitimacy.

Document Code	HBAI — 006
Document Type	Core Authority Doctrine
System Layer	3 — Solution Layer
Classification	Public
Version	1.0 — Expanded Clarity Edition
Publisher	Zyzyx.Tech — 2026
Series	Doctrine & Authority

DOCUMENT CLASSIFICATION

Document Title	Human-Bound Artificial Intelligence (HBAI)
Document Type	Core Authority Doctrine
System Layer	3 — Solution Layer
Role	Defines the conditions under which AI systems are permitted to act — establishing authority as a continuous, human-bound state rather than a static credential.
Core Function	Establishes the boundary between authorised and unauthorised action in automated systems, and defines the model through which that boundary is enforced.
Depends On	Authority at Execution (AVE-003), Human Authority Failure & Continuity Framework (HAFCF-004), Failure Justification Framework (FJF-005)
Enables	Execution-Layer Control Infrastructure (ELCI-011.5), Enforcement & Mandate Framework (EMF), Robotic Systems Governance Doctrine (RSGD)
System Position	Converts the problem definition established in preceding papers into a structured authority model with defined conditions, lifecycle states, and termination requirements.
End Condition	Systems operate only under active, validated, human-bound authority — not under assumption, inheritance, or persistence.

SYSTEM INTEGRATION STATEMENT

This document forms part of the Zyzyx Authority System, a unified governance architecture consisting of interdependent doctrine, validation models, infrastructure layers, and enforcement mechanisms.

This document must be read in conjunction with:

LTH	Letter to Humanity — A Citizen's Foundation for Automated Society
PAGF	Public Automation Governance Framework
HAFCF	Human Authority Failure & Continuity Framework
HBAI	Human-Bound Artificial Intelligence
ELCI	Execution-Layer Control Infrastructure
AFAF	Automation Failure Analysis Framework
BAF	Business Adoption Framework
EMF	Enforcement & Mandate Framework
RSGD	Robotic Systems Governance Doctrine (HBR)
FLAG	Foundational Laws of AI Governance

These documents do not operate independently. They form a single system.

> *Governance begins with understanding. It becomes enforceable only when aligned across all layers.*

NOTICE

This document defines governance doctrine only. It does not disclose implementation architecture, system design, or enforcement mechanisms.

All concepts are subject to existing or pending intellectual property protection.

This document establishes the conditions under which authority exists. It is not a design guide. It is not a technical manual. It defines the boundary between authorised and unauthorised action in automated systems.

This boundary is not theoretical. It is the point at which systems either remain under human control or operate beyond it. Where this boundary is not enforced, systems do not pause or question themselves. They continue. And that continuation is what creates unauthorised action.

This document is released for:

— Public understanding and governance alignment

— Institutional and regulatory consideration

— Policy development

It is NOT released for implementation.

ABSTRACT

Artificial intelligence systems are increasingly deployed with the capacity to initiate financial transactions, manage healthcare records, execute legal instruments, and control physical infrastructure. These are not minor interactions. These are decisions with consequence. They affect money, health, rights, access, and in some cases, physical safety.

As these systems acquire consequential authority over human affairs, the question of where authority resides, how it is established, and when it must be treated as having ended is no longer optional. It is structural. A system that can affect lives must be able to demonstrate that it was authorised to do so — not in general, not historically, but at the exact moment the action occurred.

Existing governance frameworks address behaviour. They focus on fairness, bias, safety, and transparency. These are legitimate and necessary concerns. But they do not define the most critical condition: when does a system have the right to act? Without that definition, systems continue operating after the human conditions that justified their operation have changed or no longer exist. The system is still acting — but no longer acting under valid authority.

This is not a rare failure. It is not an edge case. It is not a system defect. It is the default condition of current systems operating at scale — because authority has never been explicitly defined as a requirement that must be satisfied at execution.

> *Human-Bound Artificial Intelligence (HBAI) corrects this. It does not improve behaviour. It defines authority.*

Under HBAI, an AI system is bound to a specific human individual — the human anchor. Its authority begins with that human, exists only while conditions are valid, and ends when those conditions are no longer met. Authority does not persist. Authority does not transfer. Authority does not assume. Authority must be present — actively — at the moment of execution.

This requirement is not an optimisation. It is the minimum condition for legitimacy. This document defines that model.

SECTION I

The Structural Problem

AI systems have been given the ability to act before we defined what gives them the right to act. This is the root issue. Capability has advanced faster than the governance frameworks needed to constrain it. Systems have been deployed with the power to make consequential decisions without any requirement to demonstrate, at the moment of each decision, that they are authorised to make it.

Authority has been implied, assumed, or inherited from system access, rather than explicitly defined and maintained. A system is granted access, and access is treated as though it were authority. A system is deployed with a purpose, and that purpose is treated as though it were a continuous mandate. A system is authenticated at the start of a session, and that authentication is treated as though it validated every subsequent action.

This creates a gap between who a system appears to act for and whether it is actually authorised to act. A system can appear legitimate while operating without legitimate authority. That is the failure condition — and it is a condition that persists silently, because nothing in the system's normal operation signals that it has been crossed.

This gap is not visible during normal operation. Systems continue functioning. Outputs are produced. Actions occur. Nothing looks broken. But legitimacy is not determined by appearance. It is determined by whether authority exists at the moment of action. When that condition is not enforced, systems do not stop. They continue. And that continuation is where unauthorised action begins.

This is why the problem persists unnoticed. Systems are judged by output, not by authority. And output alone cannot confirm legitimacy. A system can produce correct-looking outputs under invalid authority. The outputs do not reveal the failure. Only the authority record does — and in most systems, no such record exists at the moment of action.

The sequence of development in modern systems has prioritised capability over legitimacy. Systems have been designed to optimise performance, efficiency, and scale before establishing the conditions under which their actions are permitted. This inversion creates a structural imbalance. In traditional human decision-making environments, authority is inherently linked to accountability. A decision is made by a person, at a specific moment, under identifiable conditions. Responsibility is traceable. In automated environments, that linkage is weakened. Systems operate continuously, often without a discrete moment where authority is re-established.

The absence of this re-establishment is not treated as a failure within current system design. It is treated as normal operation. This creates a condition where actions are derived from prior states rather than current validation, authority is inferred from system configuration rather than confirmed at execution, and responsibility becomes distributed across processes rather than anchored to a decision point. The result is not random failure. It is consistent, repeatable behaviour operating under invalid assumptions.

This structural gap produces observable outcomes across multiple domains:

Financial systems	Transactions may be executed based on prior approval conditions that no longer apply at the moment of transfer.
Healthcare systems	Automated recommendations may be generated using data that is no longer current or contextually valid.
Legal and administrative systems	Decisions may be enforced without reassessment of the conditions under which those decisions were originally justified.

In each case, the system performs correctly according to its design. The failure is not in execution. It is in the absence of authority validation at the moment of action.

Systems do not recognise this condition as an error because no rule within the system defines it as one. A system can only detect deviations from its defined parameters. If authority validation is not a defined parameter, its absence is not detectable. No alert is triggered. No correction is initiated. No pause occurs. No escalation is enforced. The system continues operating under the assumption that prior validity remains sufficient. That assumption is not evaluated. It is embedded.

The boundary between authorised and unauthorised action is not determined by whether a system produces correct outputs. It is determined by whether authority exists at the moment those outputs are generated. A system operating without authority is not partially correct. It is operating outside governance. Correct outcomes do not validate unauthorised action. They obscure it.

> *AI systems have been given the ability to act before we defined what gives them the right to act. That is the root issue.*

SECTION II

Definition of HBAI

Human-Bound Artificial Intelligence is defined as:

> ***An artificial intelligence system whose authority, operational existence, and continuity are bound exclusively and inseparably to a specific, identified human individual — the human anchor.***

This definition is restrictive by design. It does not describe what systems do. It defines what they are allowed to be. Authority under HBAI is personal — tied to a specific individual, not to a role or account. It is conditional — dependent on the current state of the human anchor and the context in which the system operates. It is time-bound — it does not persist beyond the conditions that established it. And it is non-transferable — it cannot be delegated, assigned, or passed to another system or person without re-establishing the conditions from the start.

This definition removes ambiguity at the foundation. It prevents authority from being treated as something that can exist without a clear origin, without a defined state, or without a current human condition supporting it. Without this restriction, authority becomes something that systems appear to hold, rather than something they are actually permitted to exercise. That difference is subtle in appearance, but critical in consequence.

A system that appears authorised can continue operating indefinitely. A system that is required to be authorised must continuously prove that condition. This is the shift from assumption to validation — and it is the shift that defines HBAI.

When authority is not explicitly bound in this way, it becomes distributed across systems, accounts, and processes. It is no longer clear who the system is acting for, whether that person is still valid, or whether the conditions still apply. Binding authority to a specific human removes that ambiguity. It forces every action to answer a simple question: who is this for, right now?

If that question cannot be answered clearly and immediately, authority does not exist. And if authority does not exist, action must not occur.

> *If the question 'who is this for, right now?' cannot be answered clearly and immediately, authority does not exist. And if authority does not exist, action must not occur.*

SECTION III

The Human Anchor

Every HBAI system must be bound to a single, identifiable human. This person is the human anchor. The system does not exist independently of this anchor. It does not operate for a user base, for an account, or on behalf of an organisational profile. It operates under the authority of a specific human individual — and that authority is the only basis on which it is permitted to act.

Not a username	A username is a label. It can persist after the person it represents has left, lost capacity, or been compromised.
Not an account	An account is a technical construct. It can remain active long after the authority conditions associated with it have ceased to apply.
Not a device	A device is a delivery mechanism. Possession of the device does not establish the cognitive state, intent, or voluntariness required for valid authority.

The anchor is a real person with legal identity, cognitive state, and situational awareness. These three attributes are what make the anchor's authority meaningful. Legal identity establishes accountability — there is a person who can be held responsible. Cognitive state establishes capacity — the person is able to understand what is being authorised. Situational awareness establishes context — the person knows the conditions under which the system is acting.

This distinction is essential because systems are built to operate on representations — accounts, profiles, tokens. But authority does not originate from representations. It originates from the human condition behind them. If a system is bound to an account instead of a human, the account can persist, the system can persist, but authority may no longer exist. This is how systems continue acting after authority has already failed — the account still exists, the system still runs, but the human condition that justified action is no longer valid.

The human anchor ensures that authority has a real origin, that authority can be evaluated in real terms, and that authority can be invalidated when human conditions change. Without a human anchor, systems operate in abstraction. With a human anchor, systems operate within accountability. And accountability is the condition that allows authority to exist at all.

> *Authority does not originate from representations — accounts, tokens, profiles. It originates from the human condition behind them. When that condition changes, authority changes with it.*

SECTION IV

Authority as a Dynamic State

Authority is not permanent. It is a state that must be valid at the moment an action occurs. It exists only while its conditions are satisfied — and those conditions are not fixed. They change with time, with context, with the state of the human anchor, and with the nature of the action being taken.

Authority is not something that is granted once and continues. It is something that must be true now. That is the difference between historical approval and current authority — and it is a difference that most systems do not recognise, because they were designed for persistence rather than for continuous validation.

Authority can change due to time — conditions that were valid when authority was established may have expired. It can change due to context — the situation in which an action is occurring may no longer match the conditions under which authority was granted. It can change due to human condition — the anchor may have lost capacity, been compromised, or had their intent change. It can change due to external influence — factors outside the system may have altered the validity of the original authorisation. These changes are constant. They do not require system failure to occur. They do not require visible signals. They happen naturally as part of real-world conditions.

A system that assumes authority continues is ignoring this reality. This is where most systems fail — they treat authority as static because it is easier to design for persistence than for change. But persistence is not correctness. A system that continues acting under outdated conditions is not stable. It is operating incorrectly while appearing stable.

That appearance is what makes the failure dangerous. Nothing forces a correction. Nothing signals a stop. The system continues — confidently — under conditions that no longer justify action. And the longer that continues, the further the system moves away from valid authority without recognising it.

> *A system that continues acting under outdated conditions is not stable. It is operating incorrectly while appearing stable. That appearance is what makes the failure dangerous.*

SECTION V

Authority Lifecycle

Authority moves through defined states. These states are not optional classifications — they are required for control. Without lifecycle awareness, a system has no mechanism to recognise that its authority has changed, weakened, or ceased to exist.

1 — Established	Authority has been created. The human anchor has been identified, the conditions have been defined, and the system has been permitted to begin operating.
2 — Active	Authority is valid and current. The conditions under which it was established continue to apply. The system may act.
3 — Challenged	Authority is in question. A condition has changed, or cannot be confirmed. The system must not continue acting until authority is re-established or terminated.
4 — Invalid	Authority no longer exists. The conditions required to support it have failed. Actions must cease.
5 — Terminated	Authority has formally ended. The system's operational existence under this authority is closed. No further action is permitted.

Authority is not binary. It is not simply on or off. It exists within a lifecycle that reflects real-world conditions — conditions that change continuously and that a governed system must continuously track.

Authority does not remain stable over time. It can weaken, become uncertain, become invalid, or require revalidation — and the system must recognise each of these transitions. If it does not, it cannot manage authority. It can only continue acting as if nothing has changed.

Lifecycle awareness introduces control points — defined moments at which the system must assess its authority state and decide whether to continue, pause, or stop. Without lifecycle recognition, systems skip these control points entirely. They move from active to continued operation without ever checking whether authority still exists. That means the system never has a defined moment to question itself. It simply continues. And continuation without evaluation is the mechanism through which authority failure becomes permanent.

> *Continuation without evaluation is the mechanism through which authority failure becomes permanent.*

SECTION VI

Non-Transferability

Authority cannot be transferred between individuals or systems. It is bound to a specific human, under specific conditions, at a specific moment. This is not a procedural restriction — it is a structural property of legitimate authority. Authority that can be transferred is authority that has been detached from its source, and detachment from source is the point at which accountability breaks down.

Authority is not a resource that can be handed off. It is a condition tied to a person's identity, awareness, and intent. These three attributes cannot be transferred — they are properties of the specific individual who holds authority, and they change when that individual changes. If authority appears to transfer, what has actually transferred is a representation of authority — not the condition itself.

If authority can be transferred, it loses its connection to original intent. It becomes detached from responsibility — because the person now appearing to hold it did not establish the conditions under which it was created. It becomes difficult to verify — because verification requires tracing back to a human condition, and that trace is broken at the point of transfer. A transferred authority is no longer clearly linked to the human who established it. This breaks the chain of accountability.

Systems often allow forms of implicit transfer through shared access, delegated permissions, and automated continuation. Each of these introduces ambiguity — and ambiguity is where unauthorised action becomes possible. Because once authority is no longer clearly tied to a single origin, it becomes unclear who is responsible for the action being taken. And when responsibility is unclear, enforcement becomes weak. And when enforcement is weak, systems continue acting without challenge.

> *A transferred authority is no longer clearly linked to the human who established it. This breaks the chain of accountability — and accountability is the condition that makes authority real.*

SECTION VII

Authority Must Be Validated at Execution

If authority does not persist, does not transfer, and cannot be assumed, it must be established at the point of execution. Execution is the moment at which a system produces a consequential outcome. It is the point where a decision transitions into action.

Authority cannot be inferred from prior validation states. It must be confirmed at the moment the action is initiated. Validation performed at entry, configuration, or prior approval does not establish authority for subsequent actions. These steps confirm access conditions, not execution legitimacy.

Without validation at execution, a system cannot determine whether current conditions still justify the action. It continues operating based on historical validation rather than present state. Authority must therefore be validated at execution to ensure that each action is justified under current conditions.

Execution represents the boundary between potential and outcome. Before execution, system state can be evaluated without consequence. After execution, outcomes are fixed and may not be reversible. Control must be applied at this boundary. Validation at earlier stages cannot guarantee correctness at execution due to changes in context, data, or human conditions. Execution is therefore the only point at which authority validation can reliably determine legitimacy.

Authority validation at execution requires that each consequential action is independently assessed. This establishes a model in which authority is evaluated per action, continuation depends on current conditions, and invalid conditions interrupt execution. This replaces persistent authority with condition-based operation.

> *Execution is the boundary between potential and outcome. It is the only point at which authority validation can reliably determine legitimacy.*

SECTION VIII

Systems Without Authority Control Are Not Governed

A system that does not validate authority at execution cannot be considered governed. It may meet requirements for security, reliability, or performance, but these do not establish control over action. Control requires the ability to prevent execution when authority conditions are not satisfied.

Without this capability, system behaviour is based on assumed legitimacy derived from prior validation states. Assumption does not provide control. It allows execution without confirmation. A system operating under these conditions may produce consistent outputs, but it cannot ensure that those outputs are authorised. Governance requires validation at execution. Without it, system operation is ungoverned regardless of performance characteristics.

Operational stability does not indicate control. Systems may exhibit consistent behaviour, predictable outputs, and low error rates while still operating without authority validation. These characteristics reflect system performance, not governance. Control is defined by the ability to validate and interrupt actions at execution. Without this, stability can coexist with unauthorised behaviour.

A governed system must:

Validate authority at execution	Authority conditions must be confirmed at the moment of action — not at an earlier point that conditions may have since departed from.
Interrupt actions when conditions are invalid	The system must be capable of stopping when authority cannot be confirmed. Detection without interruption is not control.
Maintain traceable justification for outcomes	Every consequential outcome must be traceable to a specific, current, human-bound authority at the moment it was produced.

Without these properties, system behaviour cannot be considered controlled.

> *Governance requires validation at execution. Without it, system operation is ungoverned — regardless of performance, stability, or apparent correctness.*

SECTION IX

Human-Bound AI — Structural Model

Human-Bound Artificial Intelligence defines a model in which authority is bound to the human at the point of execution. System actions are conditional on current authority validation rather than prior approval states. This establishes a framework in which systems operate under continuously validated human authority.

Authority originates from the human	The source of every action is traceable to a specific, identifiable human individual. Systems cannot self-originate authority.
Authority is validated at execution	Confirmation occurs at the moment of action, not at access or configuration. Prior states do not carry forward.
Authority is not treated as persistent	The system does not assume that authority established at one moment continues to apply at the next. Each action requires its own validation.
Authority is not transferred without reassessment	Any change in the human anchor or delegation requires re-establishment of authority conditions from the start.

This model introduces conditions that change how systems relate to authority. Authority is treated as dynamic — not assumed to continue from a prior state. Validation occurs at action rather than at access. Execution is dependent on current human-linked validation rather than inherited permission. And system continuation is conditional — subject to ongoing authority confirmation rather than a single initial grant.

Existing systems can operate within this model if execution is gated by authority validation. This requires interception at execution points, evaluation of current authority conditions, and interruption capability when validation fails. The model does not replace existing infrastructure. It adds a control layer at execution.

> *The HBAI model does not replace existing infrastructure. It adds the one layer that is currently missing — control at the point where consequence is created.*

SECTION X

Authority as a Governed State

Authority must be treated as a governed state rather than an assumed condition. A governed state requires continuous validation aligned with current conditions. This contrasts with systems that treat authority as persistent once established — where the original grant is assumed to remain valid indefinitely, regardless of how conditions have changed.

In a governed model, authority is re-evaluated at execution, actions depend on current validation, and outcomes are justified at the moment they occur. This defines the operational difference between governed and ungoverned systems — not in their capability, not in their performance, and not in their outputs, but in whether the condition that makes their actions legitimate is present at the moment those actions are taken.

Systems operating under governed authority do not execute actions without validation. They evaluate conditions at each decision point. They maintain alignment between action and current context. This creates traceability between authority and outcome — the ability to demonstrate, for any consequential action, that authority existed and was confirmed at the moment it occurred.

Authority is not an optional component of system design. It defines whether system actions are legitimate. A system that operates without authority validation at execution cannot confirm that its actions are justified. This condition is not limited to specific technologies. It applies across all systems that produce consequential outcomes. Until authority is validated at execution, systems will continue to operate based on prior assumptions rather than current conditions — resulting in actions that may be technically correct but not legitimately authorised.

> *Authority is not an optional component of system design. It defines whether system actions are legitimate. Without it, correctness and legitimacy are not the same thing.*

SECTION XI

Non-Compliance Outcome

Systems that do not enforce authority at execution operate without valid control. This is not a partial failure — it is a complete absence of authority governance. The system may function. It may produce outputs. It may appear correct. But without authority enforcement at execution, it cannot be called a governed system.

Non-compliance leads to unauthorised actions — decisions and outputs that cannot be traced to a valid, current, human-bound authority at the moment they occurred. It leads to unclear responsibility — because the chain from system action back to accountable human has been broken, and reconstructing it after the fact may be impossible. It leads to increased systemic risk — because each uncontrolled action creates the possibility of consequences that compound across interconnected systems. And it leads to reduced trust among the people, institutions, and regulatory bodies that depend on the system to operate within defined authority.

These are not edge outcomes. They are expected outcomes at scale. A system without authority enforcement is not mostly correct. It is operating without the condition that defines correctness. That means even correct outcomes are not reliably justified — because correctness without authority is coincidence, not control.

> *Correctness without authority is coincidence, not control. A system that produces correct outputs under invalid authority is not a governed system — it is a fortunate one.*

SECTION XII

Alignment with Existing Systems

HBAI does not replace legal or governance systems. It aligns system behaviour with them. The principles that HBAI enforces at execution are not new — they are extensions of the same conditions that legal and governance frameworks have always required of human decision-making, now applied to the automated systems that increasingly make decisions in place of humans.

Legal systems already define consent — the requirement that authority be freely given by a person with the capacity to give it. They define capacity — the requirement that the authorising person be in a state that allows them to understand what they are authorising. They define accountability — the requirement that there be a person or institution who can be held responsible for the decisions made under their authority. HBAI ensures that automated systems operate within these same conditions at the moment of action.

HBAI does not introduce new principles. It enforces existing ones at execution. The principles were already there. What was missing was the mechanism to ensure that automated systems — which now make decisions at the speed and scale that humans cannot — operate within those principles continuously, not just at the point of initial setup.

Without this alignment, systems operate outside legal expectations. Outcomes conflict with governance frameworks. Enforcement becomes reactive rather than built-in. Alignment ensures that systems do not just function — they function within accepted rules of authority.

> *HBAI does not introduce new principles. It enforces existing ones at execution — the same conditions that governance and law have always required of human decision-making.*

SECTION XIII

Final Doctrine

Authority must originate from a human. This is not a preference — it is the condition that makes authority legitimate. Systems cannot self-generate authority. Autonomous inference, probabilistic output, or system-generated determination does not constitute human authorisation and must not be treated as equivalent. The source of authority must always be traceable to a specific, identifiable person.

Authority must exist at execution — not at setup, not at login, not at the point when the task was initiated, but at the exact moment the action occurs. It must be verifiable — the system must be able to demonstrate, with a clear record, that authority was confirmed at the moment of action. It must be revocable — the system must be capable of having its authority withdrawn at any point, for any valid reason, without exception. And it must be bounded — defined by scope, context, duration, and consequence, with no action permitted outside those boundaries.

Originate from a human	Authority must have a real human source — a specific, identifiable individual who is accountable for the actions taken under their authority.
Exist at execution	Authority must be valid and confirmed at the exact moment an action occurs — not at an earlier point that may no longer reflect current conditions.
Be verifiable	Authority must be demonstrable — traceable to a specific human decision, at a specific moment, under conditions that can be reviewed.
Be revocable	Authority must be capable of being withdrawn at any time. A system that cannot be stopped does not have valid authority — it has unconstrained capability.
Be bounded	Authority must have defined limits: scope, context, duration, and consequence. Operation outside those limits is unauthorised action, regardless of outcome.

These are not recommendations. They are minimum conditions for authority to exist. If any of these conditions fail, authority does not weaken and it does not degrade. It ceases to exist. There is no partial authority — there is authority, or there is not.

This is the defining boundary — not between good systems and bad systems, but between authorised action and unauthorised action. Once that boundary is crossed, the system is no longer operating under valid authority. And every action beyond that point exists outside controlled governance — regardless of how correct those actions may appear, and regardless of how confidently the system continues to produce them.

> *There is no partial authority. Authority either exists — or it does not. When it does not, every action the system takes is unauthorised.*

FINAL STATEMENT

> **A system that cannot prove authority at the moment of action does not have the right to act.**